

## Glossary

# Glossary I

| Term                                                 | Meaning                                                                                                                                                                                                                       |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Record / Paper</b><br><b>Canonical identifier</b> | A single bibliographic entry with metadata and identifiers.<br>The highest-priority available ID (DOI > arXiv > Semantic Scholar > OpenAlex > title digest) used for de-duplication and citation resolution.                  |
| <b>Engine</b>                                        | An independent literature source adapter (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv/medRxiv, and medrxiv) with a uniform search interface and graceful skip-on-failure. |
| <b>Subfield</b>                                      | One of the 6 configurable keyword-defined buckets (Clinical Sleep, Cognition, Pharmacology, Psychiatry, Safety, and Neuroscience) into which records are classified.                                                          |
| <b>Topic</b>                                         | A latent theme from non-negative matrix factorization over the TF-IDF representation.                                                                                                                                         |
| <b>Embedding</b>                                     | A deterministic offline vector (TF-IDF → truncated SVD) for a title, abstract, or full text.                                                                                                                                  |
| <b>Hypothesis</b>                                    | One of the 6 configured claims about the topic, optionally scored by the knowledge-graph stage.                                                                                                                               |
| <b>Assertion</b>                                     | A directional (supports / contradicts / neutral) statement extracted from a record against a hypothesis, with a confidence score.                                                                                             |
| <b>Nanopublication</b><br><b>CAGR</b>                | An RDF-serialized assertion plus its provenance.<br>Compound annual growth rate of publication volume (5.48% for this corpus).                                                                                                |
| <b>Living literature review</b>                      | A synthesis that can be re-executed as the field evolves, with every number regenerable.                                                                                                                                      |